

Program9: Organize the Kubernetes Pods using the Labels and Namespaces

Step1: Create a file for namespaces.yaml

Step2: create a file as pods.yaml

Step3: apply the namespaces & pod yaml file

```
apiVersion: v1
kind: Namespace
Metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step4: verify the pods by namespaces

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
---
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step5: apply the namespaces and pods yaml file

```
Kubecttl apply -f namespaces.yaml
```

```
Kubecttl apply -f pods.yaml
```

Step6: verify pods by namespaces

```
Kubecttl get pods -n dev
```

```
Kubecttl get pods -n prod
```

Step7: check pod status

```
Kubecttl get pods
```

Step8: filter pods by label

```
Kubecttl get pods -n dev -l app=frontend
```

```
Kubecttl get pods -n prod -l app=backend
```

```

1rv24mc101_shreya@shreya-Lenovo-IdeaPad-S145-15IWL:~$ cd Deveops
1rv24mc101_shreya@shreya-Lenovo-IdeaPad-S145-15IWL:~/Deveops$ cd program9
1rv24mc101_shreya@shreya-Lenovo-IdeaPad-S145-15IWL:~/Deveops/program9$ kubectl g
et pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           42m
frontend-pod  1/1     Running   0           42m
1rv24mc101_shreya@shreya-Lenovo-IdeaPad-S145-15IWL:~/Deveops/program9$ kubectl g
et pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           42m
frontend-pod  1/1     Running   0           42m
1rv24mc101_shreya@shreya-Lenovo-IdeaPad-S145-15IWL:~/Deveops/program9$ kubectl g
et pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           42m
1rv24mc101_shreya@shreya-Lenovo-IdeaPad-S145-15IWL:~/Deveops/program9$ kubectl g
et pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           42m
1rv24mc101_shreya@shreya-Lenovo-IdeaPad-S145-15IWL:~/Deveops/program9$

```